Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2003-I-Q01] | Part A (Q1-Q5)

Answer: 839

1. The number is $720!/6 = 120 \cdot 719!$. No $n \geq 720$ works, since then $kn! \geq 720! > 720!/6$.

2. Thus $n = 719, k = 120$, giving 839.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2007-II-Q02] | Part A (Q1-Q5)

Answer: 200

1. Write $b = au, c = av$, where $u, v \geq 1$. Then $a(1 + u + v) = 100$, so a is a divisor of one hundred with $100/a \geq 3$.

2. The possible a values are $1, 2, 4, 5, 10, 20, 25$, each contributing $100/a - 2$ positive ordered pairs. Their sum is $100 + 50 + 25 + 20 + 10 + 5 + 4 - 14 = 200$.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2015-II-Q01] | Part A (Q1-Q5)

Answer: 131

1. The ratios reduce to $39/50$ and $29/25$. Therefore the integral value N must be a multiple of both 39 and 29.

2. These numbers are coprime, so the smallest value is $39 \cdot 29 = 1131$. It is attained using the original integers $1450, 975$, and its remainder is 131.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2017-I-Q02] | Part A (Q1-Q5)

Answer: 062

1. Equal remainders force $m \mid \gcd(787-702, 855-787) = \gcd(85, 68) = 17$. The divisor1 cannot leave a positive remainder, so $m = 17, r = 5$.

2. Similarly $n \mid \gcd(310, 93) = 31$, and positivity forces $n = 31, s = 9$. The sum is $17 + 31 + 5 + 9 = 62$.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2019-II-Q03] | Part A (Q1-Q5)

Answer: 096

1. As $c \mid 70$, $e \mid 72$ and $cde = 71$ prime, neither c nor e can be seventy-one. Hence $c = e = 1, d = 71$.

2. Then $ab = 70$ has eight ordered factor pairs and $fg = 72$ has twelve. Multiply to obtain 96.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2020-I-Q04] | Part A (Q1-Q5)

Answer: 093

1. If the prefix is k, the number is $10000k + 2020$. Its divisibility by k is equivalent to $k \mid 2020$.

2. As $2020 = 2^2 \cdot 5 \cdot 101$, its positive divisors are
$1, 2, 4, 5, 10, 20, 101, 202, 404, 505, 1010, 2020$. Their digit sums total 45.

3. Each of the 12 suffixes contributes 4, so the total is $45 + 48 = 93$.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2022-II-Q01] | Part A (Q1-Q5)

Answer: 154

1. Write the initial counts as $12k$ people and $5k$ adults, with k a positive integer. The final adult count is $11(12k+50)/25$.

2. Integrality requires $25 \mid 12k$, so $25 \mid k$. The smallest choice $k=25$ gives 350 total and 154 adults.

3. This is feasible: initially there are 125 adults; adding 29 adults and 21 children produces the required final counts. Hence 154 is attained.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2025-II-Q02] | Part A (Q1-Q5)

Answer: 049

1. Modulo $n+2$, substitute $n=-2$: the product becomes $3 \cdot 1 \cdot 13 = 39$. Hence the condition is exactly $(n+2) \mid 39$.

2. The divisors at least 3 are $3, 13, 39$, yielding $n = 1, 11, 37$. Their sum is 49.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2021-I-Q10] | Part B (Q6-Q10)

Answer: 059

1. For each reduced pair define $d = n - m$ and $I = 19m - 18n$. Before reduction the shift preserves I and changes the difference to $d + 1$. The reduction factor is $g = \gcd(d + 1, I)$, so the updated quantities are $(d + 1)/g, I/g$.

2. Initially $(d, I) = (1, 2002)$ with $2002 = 2 \cdot 7 \cdot 11 \cdot 13$. Cancellation successively uses factors $2, 7, 11, 13$, returning d to 1 at indices $2, 8, 18, 30$. Including the initial term gives indices $1, 2, 8, 18, 30$.

3. After I becomes 1, no reduction is possible and d grows forever. Hence the sum is 59.

Source / 来源

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2025-II-Q07] | Part B (Q6-Q10)

Answer: 237

1. $2025 = 3^4 5^2$ has 15 divisors. The LCM condition requires a selected divisor with full 3-exponent and one with full 5-exponent.

2. Of the divisors, 12 lack full 3-exponent, 10 lack full 5-exponent, and 8 lack both. Inclusion–exclusion gives $2^{15} - 2^{12} - 2^{10} + 2^8 = 27904$ successful subsets.

3. The probability is $27904/32768 = 109/128$, giving 237.

Source / 来源